

Digital and Technology solutions: Discuss how data analytics can be used to influence people

In an ever increasing data driven society, with the world generating 2.5 quintillion bytes per day¹, data analytics is more important than ever; data analytics being the process of collecting, organizing, and analyzing data in order to extract insights and make informed decisions. One market that has pioneered data analytics by harvesting the most vulnerable and sensitive data out of their users is the dating app industry. In 2021, there were 323.9 million dating app users², that's 323.9 million people entrusting these apps with their most personal data for the hope of a romantic future.

Dating apps operate matchmaking algorithms to match users with potential partners based on the analytics of user's uploaded preferences and behaviors, gathered by the app as everything you interact with when logged into the app is detected, tracked, and stored. Different dating apps utilize different algorithms, however a vast majority operate with collaborative filtering³ as a method of sorting data. As a result, this can have a profound impact on narrowing the choices a user matches with as the algorithm bases its predictions on the user's personal preferences as well as the opinion of the majority. One concern about the use of collaborative filtering for matchmaking is the potential for gender and racial bias to creep into the algorithms. This was evidently exposed through the simulation game MonsterMatch⁴ which was created to illustrate the ways collaborative filtering algorithms can exclude certain groups of users by privileging the behaviors of the majority. MonsterMatch, created by Ben Berman, Miguel Perez and Mozilla, implements the basic architecture of a dating app. You create a personal profile (from a cast of cute illustrated monsters), swipe to match with other monsters, and chat to set up dates with this imaginary monsters. However, as you swipe, the game reveals some of the more insidious consequences of dating app algorithms. The field of choice becomes extremely narrow, and you end up seeing the same monsters again and again. In the simulation, if you swipe right on a zombie and left on a mermaid, then a

new user who also swipes yes on a zombie won't see the mermaid in their queue. The MonsterMatch presentation of the dating app algorithm, demonstrates a harsh reality: Dating app users get boxed into restricted assumptions and certain profiles are routinely excluded. Therefore, it creates an echo chamber of tastes, never exposing the user to different people, different characteristics, and different personalities. This in turn amplifies a user's behavior due to the confirmation bias, the data analytics that runs these platforms feeds into a user's tendency to seek out and prefer information that supports pre-existing beliefs. As a matter of course, this algorithm inevitably leads to discrimination against minorities and marginalized groups, reproducing a pattern of human bias which only serves to deepen pre-existing divisions in the dating world.

Another way in which dating apps employ data analytics to influence users is through selling user's data to third parties in order to curate targeted advertisements. Unlike most other apps, dating apps request a lot of sensitive information from their users to build a profile. Often this includes basic information such as your name, age, gender, and date of birth. However, it can also include a lot more personal information that other apps won't collect such as, religious beliefs, sexual orientation, drugs and alcohol use, whether you have/want kids, height and weight. This can be extremely problematic if the private information shared gets sold into the wrong hands, such as extremist groups, job applications and scammers. A prevalent example of this is in 2021 the Norwegian Data Protection Authority fined Grindr, the world's most popular gay dating app, \$11.7 million⁵ for illegally disclosing users' sexual orientation and location to several advertising companies. The report highlights how these dating sites treat intimate information, like gender preference or drug habits, no differently from more innocuous information, like favorite colour. The spread of users' location and other sensitive information could present particular risks to people who use Grindr in countries, like Qatar and Pakistan, where consensual same-sex sexual acts are illegal. Furthermore, this is not the first time that Grindr has faced criticism for spreading its users' information. In 2018, another Norwegian nonprofit group found that the app had been broadcasting users' H.I.V. status to numerous companies⁶. Subsequently, these users' can be targeted by pharmaceutical

companies who now know this personal medical condition as well as putting users at risk from far-right groups who want to harm such individuals. Grindr subsequently announced that it had stopped the practice. Moreover, this surveillance system enables advertising agencies, whose names and morals are unknown to the consumer, to quietly profile individuals, target them with ads and try to sway their behavior.

In conclusion, Tinder may not get you a date but it will get your data. Consequently, as a result of the excessive amount of personal data collected by these companies, dating sites have a social and moral responsibility to protect this data and use it ethically. Finally, dating apps use data analytics to influence their users' perceptions on who they should date as the algorithms that power these apps entrap users in a cage of their own behaviors and preferences. In order to reduce/prevent these negatively biased influences on its users, dating app developers should be lawfully urged to provide users with the option to reset the algorithm by deleting their swipe history or to opt out of algorithmic matching entirely.

Bibliography

1. [Bernard Marr: How Much Data Do We Create Every Day? The Mind-Blowing Stats Everyone Should Read \(2018\), Forbes \[online\]](#)
2. [Andrej Hadji-Vasilev: 25 Online Dating Statistics & Trends in 2023 \(2022\), Cloudwards \[online\]](#)
3. [Karin Nader: Dating Through The Filters \(2021\), Cambridge University Press \[online\]](#)
4. [Benjamin Berman and Miguel Perez: MonsterMatch \(2019\) \[online\]](#)
5. [Natasha Singer and Aaron Krolik: Grindr is fined \\$11.7 million under European privacy law \(2021\), The New York Times \[online\]](#)
6. [Natasha Singer and Aaron Krolik: Grindr and OkCupid Spread Personal Details, Study says \(2021\), The New York Times \[online\]](#)